

Kashif Mahmood

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ABOUT ME

I'm Kashif Mahmood, a Software Engineering graduate specializing in artificial intelligence and data science, with research experience spanning medical imaging, agricultural AI, and applied analytics. My work includes developing a self-supervised learning framework for chest X-ray classification, independently creating and documenting a public image dataset, and delivering measurable results in applied data analytics roles. I'm continually expanding my technical range across deep learning, MLOps, and reproducible research practice.

WORK EXPERIENCE

University of Sargodha – Sargodha, Pakistan

Research Assistant

[09/2024 – Current]

- Served as lead or co-author on three research manuscripts spanning agricultural and medical imaging AI, including a self-supervised learning framework for chest X-ray classification and a transfer-learning approach to peanut disease detection, both at *Scientific Reports*.
- Independently conceived, built, and documented a public image dataset of 8,944 peanut images, including custom-built data-preparation software.
- Conducted literature reviews, experimental design, and model development across deep learning and self-supervised learning projects, in collaboration with international research partners at the Islamic University of Madinah and Princess Nourah bint Abdulrahman University.

10Pearls – Remote, Pakistan

Data Science Intern (Part-time, concurrent with Research Assistantship)

[10/2025 – 11/2025]

- Built a fully automated, end-to-end 72-hour Air Quality Index (AQI) forecasting system for Karachi's NED University region, from raw API ingestion through deployment.
- Engineered the machine learning pipeline: feature engineering, model evaluation across four algorithms via GridSearchCV, and multi-step forecasting via a MultiOutputRegressor, with all features and models versioned in Hopsworks.
- Automated the entire pipeline with dual GitHub Actions workflows for hourly data ingestion and daily model retraining, requiring no manual intervention.

BrandPur – Sargodha, Pakistan

Data Analytics Intern

[06/2024 – 08/2024]

- Analyzed customer data to understand retention and churn behavior, using Python (pandas, matplotlib, seaborn) for analysis and Power BI for dashboarding.
- Built a churn-prediction model estimating the probability of customer attrition, which informed targeted retention packages for at-risk customers.
- Contributed to an approximate 10% increase in customer retention through this data-driven retention strategy.

EDUCATION AND TRAINING

Bachelor of Science in Software Engineering

University of Sargodha, Pakistan [10/2021 – 06/2025]

City: Sargodha Country: Pakistan Website: <https://uos.edu.pk/> Final grade: CGPA 3.90 / 4.00

PUBLICATIONS

[2026]

A Deployment Risk Score Framework for Self-Supervised Chest X-Ray Classification: Calibrated Multi-Objective Evaluation Under Annotation Scarcity and Scanner Heterogeneity

- Conceived and led a comparative study of six models (contrastive and reconstruction-based self-supervised pretraining plus four supervised baselines) on a 17,988-image, six-class chest X-ray benchmark, evaluated across 10%, 20%, and 100% annotation budgets.
- Achieved macro F1 = 0.952 with SimCLR + ResNet-50, statistically comparable to the fully supervised ceiling while retaining 88.8% of full-label performance at only 10% annotation; MAE + ViT-S/16 showed the strongest robustness to synthetic perturbations and the lowest overconfidence under calibration testing.
- Introduced a Deployment Risk Score (DRS) framework translating these accuracy, robustness, and calibration trade-offs into scenario-specific model-selection guidance for clinical deployment.

Authors: Kashif Mahmood, Romana Aziz, Muhammad Ramzan, Mahwish Ilyas, Ala Saleh Alluhaidan | **Journal Name:** Scientific Reports, In Revision | **Publisher:** Nature Portfolio

[2026]

Deep Learning-Based Disease Detection in Peanut Cultivars Utilizing Transfer Learning

- Evaluated EfficientNet-B0, EfficientNet-B4, and ConvNeXt-Base as initial candidates, none of which reached state-of-the-art performance, then developed a ResNet-50 transfer-learning approach combining data augmentation, a OneCycleLR schedule, and weighted loss to handle class imbalance.
- Achieved 97.28% accuracy on a 1,720-image benchmark of healthy and diseased groundnut leaf images, surpassing the previous best published result of 96.51% on the same dataset.
- Established a new baseline for automated groundnut leaf disease detection, demonstrating the feasibility of AI-based tools for sustainable agricultural practice.

Authors: Kashif Mahmood, Qaisar Abbas, Ume Habiba, Mahwish Ilyas, Turki Alghamdi, Hani Almoamari, Muhammad Ramzan | **Journal Name:** Scientific Reports, Under Review | **Publisher:** Nature Portfolio

[2026]

A Systematic Review of Visual Computing Technologies for Intelligent Peanut Agriculture

- Reviewed deep learning, machine learning, and hybrid approaches across four core tasks in peanut agriculture: variety classification, maturity assessment, disease detection, and pest recognition.
- Compared benchmark datasets and evaluation metrics across the literature, identifying dataset scale, model complexity, and field adaptability as the primary open challenges.
- Proposed future research directions, including lightweight visual models, multi-task integrated systems, and soil-penetrating sensing technologies.

Authors: Ume Habiba, Nabeel Abbas, Kashif Mahmood, Mahwish Ilyas, Muhammad Summair Raza, Muhammad Ramzan | **Journal Name:** USJICT, Under Review | **Publisher:** University of Sindh, Jamshoro

RESEARCH OUTPUTS

PakGroundnut-4V: A Multi-Condition Image Dataset for Pakistani Groundnut (*Arachis hypogaea* L.) Variety Classification

- Created a dataset of 8,944 images across four Pakistani peanut varieties (verified by the Barani Research Institute, Chakwal), captured across five backgrounds, four lighting conditions, and five quantity levels.
- Collected morphometric measurements (length, width, weight, seeds per pod, shell thickness) using a digital caliper and weighing scale, and published complete reproducibility documentation (CITATION.cff, data card, license, README).
- Publicly available on Zenodo: <https://doi.org/10.5281/zenodo.22014996>.

Image Dataset Cropper – Peanut Dataset Preparation Tool

- Built a full-stack local web application (Python 3.11, FastAPI, Pillow, React 18, Vite) to batch-crop composite dataset photographs into individual ML-ready images, solving the bottleneck behind preparing PakGroundnut-4V.
- Designed a custom HTML5 Canvas multi-region crop editor supporting unlimited simultaneous crop regions with drag, resize, and one-click batch-apply across an entire folder.
- Engineered for memory efficiency at scale: chunked uploads, generator-based streaming ZIP export, and disk-cached thumbnails, enabling processing of multi-gigabyte datasets with no size limit.
- Publicly available on GitHub: <https://github.com/Kashif-Mahmood007/image-dataset-cropper>

PROJECTS

PeanutInfo – Smart Agriculture System (Final Year Project)

- Built a three-module deep learning system for peanut variety classification, pest identification, and disease detection, combining a custom-collected classification dataset with two public datasets for the pest and disease modules.
- Evaluated ResNet-50 and EfficientNet-B0/B4 before achieving state-of-the-art performance with ConvNeXt (Tiny, Small, Base variants), exceeding 98% accuracy across modules.
- Deployed a full-stack web platform (HTML/CSS/Bootstrap/JavaScript frontend, PHP/MySQL backend, Flask API) for real-time image-based predictions.

PipeSense – Pipeline Condition Prediction System

- Built a regression model predicting gas and water pipeline condition from physical and environmental features, including size, diameter, material, strength, pressure, material loss, and temperature.
- Applied a Yeo-Johnson transformation, feature scaling, and ordinal and one-hot encoding within a scikit-learn Pipeline and ColumnTransformer for reproducible preprocessing.
- Achieved 97.4% accuracy, validated via 5-fold cross-validation.

Automotive Listings Scraper

- Built a Python and BeautifulSoup web scraper extracting 24,000+ structured vehicle listings (price, mileage, fuel type, transmission, engine capacity) from a local automotive platform.
- Implemented pagination handling and modular, maintainable code to scrape reliably at scale across multiple pages.
- Cleaned and exported the dataset with pandas into a CSV for market analysis and ML applications.

TECHNICAL AND SOFT SKILLS

Programming & Core Technologies:

C++, JavaScript, PHP, Python, MySQL, Pandas, NumPy, Scikit-learn, PyTorch, Matplotlib, Seaborn, BeautifulSoup.

Tools & Platforms

Jupyter Notebook, Power BI, MySQL Workbench, Git & GitHub, JIRA, Visual Studio Code, Microsoft Project, Hopsworks, GitHub Actions, CI/CD Pipeline, Streamlit

Soft Skills

Leadership, Communication, Critical Thinking, Problem Solving, Story Telling

LEADERSHIP AND COMMUNITY ENGAGEMENT

Vice President – Technical Wing, SE Spectrum

- Coordinated technical events and workshops within the Department, and delivered seminars including *"Bridging the Gap Between Data and Business Goals," "Data-Driven Decisions for Startups,"* and *"Programmatic Thinking."*
- Mentored peers in technical domains, encouraging participation in research and practical projects.

Head – Data Science Group, Techive Society (University of Sargodha)

- Led the faculty's Data Science group, planning seminars and learning sessions focused on AI, ML, and data analytics.
- Supported more than 100 junior students through workshops, technical sessions, and informal mentorship in AI, ML, and data analytics tools.

HONOURS AND CERTIFICATIONS

Awards & Recognition

- 3rd Position, Annual University Sports Gala – Badminton (representing the Department of Software Engineering)
- Champion, Inter-Departmental Badminton (Solo & Doubles) — qualifying selection event for the Annual Sports Gala
- Certificates of Appreciation for seminars delivered on data-business alignment, startup analytics, and problem-solving

Certifications

- Completed 6 courses in data science, Python, and analytics (Udemy, Coursera)

[Certificates available upon request](#)